Ass-3 :

Problem 3 :

Probability current: $J(x,t) = \frac{-\hbar}{2m} i \left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right]$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot J = 0$$

Problem 5 :

i) $(1+i)\,|z;+\rangle - (1+i\sqrt{3})\,|z;-\rangle$

Normalised spin state at direction $n(\theta,\phi)$ $\Rightarrow$ $\boxed{|n;+\rangle = \cos\frac{\theta}{2} \cdot |+\rangle + e^{i\phi} \sin\frac{\theta}{2}\, |-\rangle}$

$$C_1 = 1+i \quad = \cos\theta/2$$
$$C_2 = 1+i\sqrt{3} \quad = e^{i\phi} \sin\theta/2$$

$|C_1|^2 = 2; \; C_2^2 = 4 \quad \Rightarrow \langle \psi | \psi \rangle = 6$

$$N = \frac{1}{\sqrt{6}}$$

$|\psi\rangle = \frac{1}{\ } \Big(1+i \quad 0 \Big)$

$$\frac{\cdots}{\sqrt{6}}\begin{pmatrix} \cdots & \cdots \\ 0 & 1+i\sqrt{3} \end{pmatrix}$$

$\boxed{\text{To get direction solve for } \theta \text{ \& } \phi}$

Promblem 6 :

$n \cdot n' = \cos\gamma$, To show: $|\langle n';+|n;+\rangle|^2 = \cos^2\frac{\gamma}{2}$

$$|n;+\rangle = \cos\frac{\theta}{2}|+\rangle + e^{i\phi}\sin\frac{\theta}{2}|-\rangle$$

$$\langle n|n'\rangle = \left\langle\left(\cos\frac{\theta}{2}|+\rangle + e^{i\phi}\sin\frac{\theta}{2}|-\rangle\right)\middle|\left(\cos\frac{\theta'}{2}|+\rangle + e^{i\phi'}\sin\frac{\theta'}{2}|-\rangle\right)\right\rangle$$

$$\cos^2\frac{\gamma}{2} = \frac{\cos\gamma + 1}{2}$$

$$\langle n|n'\rangle = \cos\frac{\theta}{2}\cos\frac{\theta'}{2} + e^{i(\phi+\phi')}\sin\frac{\theta}{2}\sin\frac{\theta'}{2}$$

Ass 4 :

Promblem 1 :

$$\psi_0 = \left(\frac{\alpha}{\pi}\right)^{1/4} \cdot e^{-\alpha x^2/2} \quad ; \quad \alpha = \frac{m\omega_0}{2} \; , \; \omega_0^2 = \frac{k}{m}$$

$$\text{SHO} : \quad \hat{H} = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2$$

$$E_0 = \frac{1}{2}\hbar\omega_0 \; , \; V(x) = \frac{1}{2} m\omega^2 x^2$$

$$E = V(x) \longrightarrow \frac{1}{2}\hbar\omega_0 = \frac{1}{2} m\omega_0^2 x^2$$

$$x^2 = \frac{\hbar}{m\omega_0}$$

Turning points: $x_p = \pm \sqrt{\dfrac{\hbar}{m\omega_0}}$

integrate outside x_p

$$P = \int_{-\infty}^{-x_p} |\psi_0|^2 dx + \int_{x_p}^{\infty} |\psi_0|^2 dx$$

Problem 2 :

i) $[a, (a^\dagger)^n]$

$$\boxed{N = a^\dagger a}$$

$$a\,|n\rangle = \sqrt{n}\,|n-1\rangle$$

$$a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

$$N|n\rangle = n|n\rangle$$

$$[N,a] = -a \quad ; \quad [N,a^\dagger] = a^\dagger \quad ; \quad [a,a^\dagger] = 1$$

$$x = \sqrt{\frac{\hbar}{2m\omega}}\,(a^\dagger + a)$$

$$p = i\sqrt{\frac{m\omega\hbar}{2}}\,(a^\dagger - a)$$

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3 $\vec{B} = B_0 \cos\omega t \cdot \vec{e}_z$

$$\mathcal{U}(t) = \exp\left\{-\frac{i}{\hbar}\int_0^t H\,dt\right\}$$

$$H = -\gamma B S_z = -\gamma S_z B_0 \cos\omega t$$

$$\mathcal{U}(t) = \exp\left\{\frac{+i}{\hbar}\,\frac{\lambda B_0}{\omega}\,\sin\omega t\ \underbrace{S_z}_{\frac{\hbar}{2}\sigma_z}\right\}$$

Spin is in S_x initially

$\hookrightarrow |\psi(0)\rangle = |S_x;+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$

$$|\psi(t)\rangle = \mathcal{U}(t)|\psi(0)\rangle = \begin{pmatrix} e^{i\delta(t)} & 0 \\ 0 & e^{i\delta(t)} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|n;+\rangle = \begin{pmatrix} \cos\theta/2 \\ e^{i\phi}\sin\theta/2 \end{pmatrix}$$

$$|\psi(t)\rangle = \begin{pmatrix} e^{i\delta(t)} & 0 \\ 0 & e^{i\delta(t)} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S_n = -\frac{\hbar}{2} \longrightarrow |S_n;-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Prob : $|\langle S_n;-|\psi(t)\rangle|^2$

Full Flip $\Rightarrow$ $\boxed{P(S_n = -\hbar/2) = 1}$

Evolution of any op $\hat{A}$ in H-dynamics

$$\frac{d\hat{A}_H}{dt} = \frac{1}{i\hbar}[\hat{A}_H, \hat{H}]$$

$[S\ S]\quad i\hbar S$

$[S_x, S_y] = i\hbar S_z$

$y \quad z$

$z \quad x$

For time ind $\hat{H}$

$$\frac{dS_z}{dt} = \frac{1}{i\hbar}[S_z, H]$$

$$= \frac{1}{i\hbar}[S_z, -\lambda B S_z] = 0$$

for S_x: $$\frac{dS_x}{dt} = \frac{1}{i\hbar}[S_z, -\lambda B S_x]$$

$$= \frac{-\lambda B}{\cancel{i\hbar}} \cancel{i\hbar} S_y = -\lambda B S_y$$